

Quan Huynh

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EDUCATION

George Mason University

(Expected) May 2028

Bachelor of Computer Science GPA: 3.94/4.0

Coursework: Data Structures and Algorithms, Object-Oriented Programming (OOP), Web Development, Databases, Computer Systems, Artificial Intelligence, Cloud Computing, Computer Networks

EXPERIENCE

Deepiri — AI R&D Collective

May 2026 – Present

AI Software Engineer Intern (Renderflow Studio Team)

- Built AI-assisted video editing workflows for **Renderflow Studio** using **Tauri**, **Rust**, **TypeScript/Vite**, and **FastAPI**, syncing **3 core entities** across project state, timeline metadata, and AI generation jobs
- Implemented AI job orchestration in **TypeScript** by binding tasks to projects, wiring **Accept/Reject** actions to **REST APIs**, and replacing polling with **Server-Sent Events (SSE)**, reducing job refresh latency by **60%**
- Developed video preview and active-clip scrubbing by integrating a custom **Rust** timeline engine with **FFmpeg**-based proxy frame extraction and native video streaming, reducing p95 preview startup latency to under **500ms**
- Built backend integration layer for a **35+ developer** cross-university codebase, standardizing **FastAPI** service contracts, Dockerized development workflows, and CI/CD validation across **Rust**, **Python**, and **TypeScript**

George Mason University

Sep 2025 – Present

Software Engineering Project Member

- Built cloud deployment template for club web projects using **Docker**, **AWS EC2**, and **GitHub Actions**, standardizing backend setup, environment variables, and CI/CD workflows for **5+ student projects**
- Developed internal project starter kit with **Next.js**, **Supabase**, **PostgreSQL**, and **Vercel**, enabling club members to launch full-stack apps faster with reusable authentication, storage, and API routing modules

Brothers Bistro

Nov 2024 – Present

Restaurant Server

- Worked efficiently in a fast-paced customer-facing environment, managing multiple requests, resolving service issues, and coordinating with team members while maintaining accuracy and professionalism
- Handled customer orders, table service, and beverage preparation while developing strong communication, prioritization, and problem-solving skills transferable to cross-functional engineering teams

PROJECTS

PaceTrack | *React.js, Node.js, Express.js, MongoDB, Socket.io, AWS S3, Mapbox GL JS*

May 2025 – Present

- Built full-stack running tracker with **React.js**, **TypeScript**, **Node.js**, and **Express.js**, supporting authentication, activity logging, club membership, and ownership-scoped authorization across **3 core data models**
- Implemented real-time club leaderboard system using **Socket.io** rooms, broadcasting saved runs to connected members and keeping rankings synchronized across concurrent client sessions with low-latency updates
- Designed GPS route ingestion pipeline with **AWS S3** presigned uploads, custom **GPX** parsing, haversine distance calculation, and elevation gain computation for server-side run analytics and route summaries
- Rendered interactive route maps with **Mapbox GL JS** and stored **GeoJSON** LineStrings, adding auto-fitted bounds, start/end markers, and live-run persistence from timestamped GPS coordinates for map playback

TradingLab | *React.js, TypeScript, NestJS, MongoDB, Redis, OpenAI API, Docker*

Apr 2026 – Present

- Built backend price service and portfolio backtesting APIs in **NestJS** and **TypeScript**, fetching historical market data from **CoinGecko** and **AlphaVantage** to support **BTC**, **ETH**, **AAPL**, and **TSLA** simulations
- Implemented multi-asset DCA portfolio workflow with **MongoDB**, weight-allocation validation, and reusable **React** asset components, enabling users to split recurring investments across assets with weights summing to **100%**
- Integrated AI advisor infrastructure using **OpenAI API**, **Server-Sent Events (SSE)**, and **Redis**-backed rate limiting, streaming real-time investment insights from portfolio diagnostics and backtest context

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, TypeScript, JavaScript, Go, C, Bash, HTML/CSS, SQL

Frameworks/Tools: Spring Boot, Flask, FastAPI, .NET, React.js, Next.js, Node.js, Express, Tailwind, Redux, gRPC

Databases: PostgreSQL, MongoDB, Redis, MySQL, DynamoDB, Supabase

Cloud/Infra: AWS (ECS, S3, RDS, Lambda), GCP, Docker, Kubernetes, Nginx, GitHub Actions, Terraform, Linux

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, LangChain, OpenAI API